

$$\begin{aligned}
& K g(0) \left[\ln\left(\frac{\pi}{2}\right) - \frac{3}{2} \ln N \right] - \frac{K}{2\pi} \int_{-\infty}^{\infty} h(t) \left[\frac{\Gamma'(\frac{1}{2} + it)}{\Gamma(\frac{1}{2} + it)} + \frac{\Gamma'(1 + it)}{\Gamma(1 + it)} \right] dt \\
& + 2K \sum_{n \equiv \varepsilon \pmod{N}} \frac{\Lambda(n)}{n} g(2 \ln n) + \frac{1}{8} h(0) 2^r [\phi(N) - 2] \\
& + 2^{r-1} \left[\sum_{p|N} F(N_p^{-1}) \Lambda(p) g(0) + 2 \sum_{p|N} \sum_{\substack{n \in \langle p \rangle \\ n \equiv \varepsilon \pmod{\frac{N}{p}}}} F(N_p^{-1}) \frac{\Lambda(n)}{n} g(2 \ln n) \right] . \quad \boxplus
\end{aligned}$$

It is understood that:

$$F(n) = \begin{cases} 1, & n=1 \\ \frac{1}{2} \phi(n), & n \neq 1 \end{cases}, \quad \varepsilon = \pm 1, \quad N = p_1 \cdots p_r, \quad K = \frac{1}{2} \phi(N) 2^r.$$

These results correspond to Hejhal[2, page 476 middle]. We might also mention that:

$$\begin{aligned}
(6.10) \quad \text{Tr} [\Xi'(s) \Xi(1-s)] &= K \left[2 \ln \pi - \frac{\Gamma'(s)}{\Gamma(s)} - \frac{\Gamma'(1-s)}{\Gamma(1-s)} - 3 \ln N \right] \\
&+ \frac{2K}{\phi(N)} \sum_{\substack{+ \\ \mathcal{N}_1}} \sum_{p|K_1} \ln p \frac{1 - p^{-2}}{(1 - \mathcal{N}_1 p^{-2s})(1 - \overline{\mathcal{N}}_1 p^{2s-2})} \\
&- \frac{4K}{\phi(N)} \sum_{\substack{+ \\ \mathcal{Y} \pmod{N}}} \left[\frac{L'(2s, \mathcal{Y})}{L(2s, \mathcal{Y})} + \frac{L'(2-2s, \overline{\mathcal{Y}})}{L(2-2s, \overline{\mathcal{Y}})} \right].
\end{aligned}$$

The analog of 511(2.12) is obvious. Cf. 566 line 5. Needless to say: $M_e = 0$.

7. Some remarks about Hecke groups.

Classically speaking: the Hecke group $G_N^{\boxplus}$ is that subgroup of $\text{PSL}(2, \mathbb{R})$ generated by

$$(7.1) \quad E \quad \text{and} \quad S^\lambda$$

where $\lambda = 2\cos\frac{\pi}{N}$. It is understood that N is a positive integer ≥ 3 . We can also allow $N = \infty$. The standard fundamental region for G_N is $\{z \in H: |z| > 1, |x| < \frac{1}{2}\lambda\}$. The defining relations are $E^2 = I$ and $(ES^\lambda)^N = I$.

As basic references we cite: Hecke[1, pp.591-643], Lehner[1, pp.227, 228, 234], and Rosen[1].

Note that G_∞ corresponds to the theta group. Cf. 524 line 6. The case $N = 3$ corresponds to $PSL(2, \mathbb{Z})$. Cf. section 2.

For our purposes: it is best to define

$$(7.2) \quad \mathcal{Y}_N = a\left(\frac{1}{\sqrt{\lambda}}\right) G_N a(\sqrt{\lambda}) .$$

This group has generators

$$(7.3) \quad Q_N \quad \text{and} \quad S$$

where $Q_N = \begin{pmatrix} 0 & -\lambda^{-1} \\ \lambda & 0 \end{pmatrix}$. The new fundamental region is $\mathcal{D}_N = \{z \in H: |x| < \frac{1}{2}, |z| > \frac{1}{\lambda}\}$.

Throughout this section we assume that $m = 0$, $\chi_0 \equiv 1$. Cf. page 338. Let $E_N(z; s)$ be the Eisenstein series on $\mathcal{Y}_N$ associated with $i\infty$. This notation can also be used when $N = \infty$. We write:

$$(7.4) \quad E_N(z; s) = y^s + \varphi_N(s) y^{1-s} + O(e^{-2\pi y}) \quad \text{for } y \geq 2 .$$

There should be no danger of confusion with chapter 6 proposition 8.6. By 281(e):

$$(7.5) \quad \varphi_N(s) = \sqrt{\pi} \frac{\Gamma(s - \frac{1}{2})}{\Gamma(s)} \sum_{\substack{w_{00} \in [s] \backslash \mathcal{Y}_N / [s] \\ c(w_{00}) \neq 0}} \frac{1}{|c|^{2s}} \quad \text{for } s \in \mathcal{R} .$$

As in chapter 6: $\mathcal{R} = \{s \in \mathbb{C}: \operatorname{Re}(s) > 1\}$.

LEMMA 7.1. Given $\mathcal{H} > 0$, $0 < R < 1$, $z_0 \in H$, and $s_0 \in \mathbb{C}$. Let $f(z;s)$ be a complex-valued function on

$$\mathcal{M} = \left\{ \left| \frac{z - z_0}{z - \bar{z}_0} \right| < R \right\} \times \{ |s - s_0| < \mathcal{H} \} \quad .$$

Suppose that:

- (i) $f(z;s)$ is a C^2 eigenfunction of D with eigenvalue $s(1-s)$ for EACH s ;
- (ii) $f(z;s)$ is holomorphic with respect to s for EACH z ;
- (iii) $|f(z;s)| \leq 1$.

Then:

- (a) $f(z;s)$ is C^∞ with respect to both variables simultaneously ;
- (b) the first derivatives of $f(z;s)$ are bounded on compact subsets of $\mathcal{M}$ by constants which depend SOLELY on the geometry [and not on f] .

Proof. Return to chapter 6 proposition 4.7. Review 343(★) . We can now express $f(z;s)$ in the form:

$$(7.6) \quad f = \sum_{n=-\infty}^{\infty} A_n(s) \frac{r^{|n|}}{\Gamma(1+|n|)} F(1-s, s; 1+|n|; \frac{r^2}{r^2-1}) e^{in\theta} \quad .$$

The coefficients are analytic functions of s . Cf. Hille[1, page 257] and 21 line 6 with sufficiently small r . The implied constant in 21 line 13 will depend continuously on s . Cf. theorem A1 on page 614 . It follows that:

$$(7.7) \quad A_n(s) = O \left[\frac{\Gamma(1+|n|)}{(R-\varepsilon)^{|n|}} \right]$$

with an implied constant depending solely on $R, \mathcal{H}, z_0, s_0, \varepsilon$. To finish the proof, we apply the complex variable trick mentioned in remark 4.4. Cf. page 16 . Standard Cauchy estimates take care of the 1st derivatives. ■

There is an obvious extension of (7.7) when hypothesis (ii) is deleted. [■] This remark leads to an IMMEDIATE generalization of assertion (b). We omit its explicit formulation.

LEMMA 7.2. For each $\varepsilon > 0$, there exists a positive constant $C(\varepsilon)$ such that

$$(7.8) \quad \sum_{T \in \mathcal{H}_N} \frac{(\operatorname{Im} Tz)^{1+\varepsilon}}{[1+|Tz|]^{2+2\varepsilon}} \leq C(\varepsilon) \quad \text{for every } z \in H.$$

In particular: $C(\varepsilon)$ does NOT depend upon N .

Proof. Return to chapter 6 proposition 5.1. Repeat the old proof with $\mathcal{S} = \{ |z - 2i| < 10^{-6} \}$. Consult page 28 line 5. [■]

LEMMA 7.3. For each $\varepsilon > 0$, there exists a positive constant $C_1(\varepsilon)$ such that

$$(7.9) \quad 0 \leq \varphi_N(1+\varepsilon) \leq C_1(\varepsilon).$$

Proof. Let E be a tiny disk centered at i . Take $N \geq 4$ and repeat the argument on page 63. Write $C(\delta)$ in place of $C_1(\Gamma, \delta)$. Since $\lambda \geq \sqrt{2}$ the desired estimate follows immediately. [■]

LEMMA 7.4. For each $\varepsilon > 0$, there exists a positive constant $C_2(\varepsilon)$ such that

$$(7.10) \quad \operatorname{Im} T(z) \leq C_2(\varepsilon)$$

whenever $T \in \mathcal{H}_N$, $c(T) \neq 0$, and $\operatorname{Im}(z) \geq \varepsilon$.

Proof. Write T in the form $S^\alpha W_{00} S^\beta$. Observe that $c(T) = c(W_{00})$ and that

$$\operatorname{Im} T(z) = \frac{y}{|cz+d|^2} = \frac{y}{(cx+d)^2 + c^2 y^2} \leq \frac{1}{c^2 y} \leq \frac{1}{c^2 \varepsilon}.$$

Equation (7.9) shows that $|c(W_{00})|$ has a positive lower bound independent of N . ■

LEMMA 7.5. Let $s = \sigma + it$. For each $\varepsilon > 0$, there exists a positive constant $C_5(\varepsilon)$ such that

$$(7.11) \quad |E_N(z; s) - y^s| \leq C_5(\varepsilon)^\sigma$$

whenever $\sigma \geq 1 + \varepsilon$ and $z \in \mathcal{D}_N \cup \{y \geq \varepsilon\}$.

Proof. For fixed z , we can always adjust W_0 so that $|\operatorname{Re} W_0(z)| \leq \frac{1}{2}$. Consider the position of $W_0(z)$ when $W_0 \notin [S]$. The case $z \in \mathcal{D}_N$ is trivial. The other possibility is handled by means of lemma 7.4. In either case: $|W_0(z)| \leq C_3(\varepsilon)$. Lemma 7.2 will now yield

$$\left[\frac{1}{1 + C_3(\varepsilon)} \right]^{z + 2\varepsilon} \sum_{W_0 \notin [S]} (\operatorname{Im} W_0 z)^{1 + \varepsilon} \leq C(\varepsilon).$$

Let $C_4(\varepsilon) = \max[C(\varepsilon), C_3(\varepsilon)]$. A moment's thought shows that

$$\sum_{W_0 \notin [S]} (\operatorname{Im} W_0 z)^\sigma \leq [1 + C_4(\varepsilon)]^{\sigma + 2 + \varepsilon} \leq [1 + C_4(\varepsilon)]^{3\sigma}.$$

Take $C_5(\varepsilon) = [1 + C_4(\varepsilon)]^3$. The desired conclusion follows immediately. ■

PROPOSITION 7.6. Define $\mathcal{R}$ as on page 569. Then:

$$E_N(z; s) \rightrightarrows E_\infty(z; s) \quad \text{and} \quad \varphi_N(s) \rightrightarrows \varphi_\infty(s)$$

on compact subsets of $H \times \mathcal{R}$. Cf. page 803 line 4.

Proof. It suffices to consider $E_N(z;s)$. The result for φ_N will THEN follow by a trivial integration.

To get started: note that lemma 7.5 implies that E_N is uniformly bounded on compact subsets of $H \times \mathcal{K}$. Lemma 7.1 will NOW ensure that $E_N(z;s)$ is uniformly equicontinuous on $H \times \mathcal{K}$ compacta. By virtue of this remark: it's enough to prove POINTWISE convergence.

To establish this convergence: we apply a normal families argument. Cf. Ahlfors[1,pp.210-219], Golusin[1,pp.10-14], and Kelley[1,pp.233-234].

Let $\mathcal{J}$ be ANY subsequence of $\{3 \leq N < \infty\}$ with respect to which $E_N(z;s)$ converges uniformly on compacta. Let $F(z;s)$ be the limit function. It is easily seen that:

- (i) $F(z;s)$ is continuous ;
- (ii) $F(z;s)$ is analytic in s ;
- (iii) $F(z;s)$ is C^∞ with respect to z ;
- (iv) $D_z F + s(1-s)F = 0$;
- (v) $F(z+1;s) \equiv F(z;s)$;
- (vi) $F(Q_\infty z;s) \equiv F(z;s)$.

Cf. lemma 7.1 and chapter 6 proposition 4.7. To prove (vi), we note that

$$F(Q_\infty z;s) = F(Q_\infty z;s) - E_N(Q_\infty z;s) + [E_N(Q_\infty z;s) - E_N(z;s)] + E_N(z;s) .$$

The bracketed term is handled by equicontinuity. Needless to say: $N \in \mathcal{J}$.

The CRUCIAL ESTIMATE will now appear when we take a limit in lemma 7.5. Fix any $\varepsilon > 0$. The open sets $\mathcal{D}_N$ tend to $\mathcal{D}_\infty$ in an obvious fashion $(\star)$. This GEOMETRIC CONVERGENCE shows that

$(\star)$ In fact $\mathcal{D}_\infty = \bigcup \mathcal{D}_N$.

$$(7.12) \quad |F(z; s) - y^s| \leq C_5(\varepsilon)^{\sigma} \quad \text{for EACH } z \in \mathcal{H}_{\infty}$$

when $\sigma \geq 1 + \varepsilon$. By continuity the same result holds for $\partial \mathcal{H}_{\infty}$.

Choose any $\xi \in \mathcal{R}$ with $\operatorname{Re}(\xi) \geq 1 + \varepsilon$. Define:

$$(7.13) \quad u(z) = F(z; \xi) - E_{\infty}(z; \xi) \quad .$$

This function is an automorphic eigenfunction on $\mathcal{H}_{\infty} \setminus \mathbb{H}$. In addition:

$$(7.14) \quad u(z) = O(1) \quad \text{for } z \in \mathcal{H}_{\infty} \cup \partial \mathcal{H}_{\infty} \quad .$$

Cf. (7.12) and lemma 7.5 for $\mathcal{H}_{\infty}$.

It follows that $u \in \mathcal{L}_2[\xi(1-\xi)]$. By page 282 item [2] : $u(z)$ must vanish identically. Since F is independent of $\mathcal{V}$, we are done. ■

This method of proof seems rather important. One naturally thinks of the Caratheodory kernel theorem. Cf. Hejhal[6,8] and Golusin[1, pp.46,196].

We are also reminded of Siegel[5, pp.171-172].

LEMMA 7.7. For each $\varepsilon > 0$, there exists a positive constant $C_6(\varepsilon)$ such that

$$(7.15) \quad |\varphi_N(s)| \leq C_6(\varepsilon)$$

whenever $\frac{1}{2} \leq \sigma \leq \frac{3}{2}$ and $|t| \geq \varepsilon$.

Proof. Repeat the derivation of 155(12.2) with $B = 10$ when $N < \infty$. Cf. 527(3.1) and 508(2.2) for $\mathcal{H}_{\infty}$. ■

PROPOSITION 7.8. Let $\mathcal{R}_1$ denote the half plane $\{\operatorname{Re}(s) > \frac{1}{2}\}$ with a slit along $[\frac{1}{2}, 1]$. Then $\varphi_N(s) \Rightarrow \varphi_\infty(s)$ on $\mathcal{R}_1$ compacta.

Proof. This follows from Vitali's theorem. Cf. Golusin[1, pp.13-14] and Hille[1, pp. 251-255]. Proposition 7.6 and lemma 7.7 play a crucial role. ■

LEMMA 7.9. Let A denote the rectangle $[0, 1] \times [-1, 1]$. There exists a positive constant C_7 such that

$$(7.16) \quad \sup_{0 \leq x \leq \frac{1}{10}} \int_{-\frac{1}{2}}^{\frac{1}{2}} \ln \frac{1}{|f(x+iy)|} dy \leq C_7 \ln \frac{1}{|f(\frac{1}{2})|}$$

for EVERY holomorphic function on A whose range is contained in $\{0 < |w| \leq 1\}$.

Proof. Take $0 \leq h \leq \frac{1}{10}$ and $A(h) = [h, 1] \times [-1, 1]$. Let $g(z; z_0; h)$ be the Green's function for $A(h)$. Note that:

$$\ln \frac{1}{|f(\frac{1}{2})|} = \frac{1}{2\pi} \int_{\partial A(h)} \ln \frac{1}{|f(z)|} \frac{\partial g}{\partial n}(z; \frac{1}{2}; h) |dz| \geq \frac{1}{2\pi} \int_{-\frac{1}{2}}^{\frac{1}{2}} \ln \frac{1}{|f(h+iy)|} \frac{\partial g}{\partial x}(h+iy; \frac{1}{2}; h) dy.$$

By relating $g(z; z_0; h)$ to an OBVIOUS conformal mapping and applying the Schwarz reflection principle, we discover that:

$$(7.17) \quad \inf_{\substack{0 \leq h \leq \frac{1}{10} \\ |y| \leq \frac{1}{2}}} \frac{\partial g}{\partial x}(h+iy; \frac{1}{2}; h) > 0.$$

Cf. Ahlfors[1, pp.225-227] and Hille[1, pp.399-401]. The desired estimate follows immediately. ■

LEMMA 7.10. Let $f(z)$ be holomorphic on $\{|z| < 1\}$. Suppose that

$$\iint_{|z| < 1} \ln^+ |f(z)| \, dx \, dy \leq M.$$

Then:

$$(7.18) \quad \ln^+ |f(re^{i\theta})| \leq \frac{2M}{(1-r)^2} \quad \text{for } 0 \leq r < 1.$$

Proof. The function $\ln^+ |f(z)|$ is subharmonic. Hence:

$$(7.19) \quad \ln^+ |f(re^{i\theta})| \leq \frac{1}{2\pi} \int_0^{2\pi} \frac{R^2 - r^2}{R^2 + r^2 - 2Rr \cos(\theta - \phi)} \ln^+ |f(Re^{i\phi})| \, d\phi \quad \text{for } r < R.$$

Cf. Ahlfors [1, pp. 237-239] and Golusin [1, pp. 280-282]. An integration with respect to θ yields:

$$(7.20) \quad \int_0^{2\pi} \ln^+ |f(re^{i\theta})| \, d\theta \leq \int_0^{2\pi} \ln^+ |f(Re^{i\phi})| \, d\phi.$$

This shows that

$$M \geq \int_0^1 \int_0^{2\pi} r \ln^+ |f(re^{i\theta})| \, d\theta \, dr \geq \int_{\frac{1}{2}}^1 r \, dr \cdot \int_0^{2\pi} \ln^+ |f(\xi e^{i\theta})| \, d\theta.$$

In particular:

$$(7.21) \quad \int_0^{2\pi} \ln^+ |f(Re^{i\theta})| \, d\theta \leq \frac{2M}{1-R^2} \quad \text{for } 0 < R < 1.$$

Take $R = \frac{1+r}{2}$ and apply (7.19). We conclude that

$$\begin{aligned}
 |n^+ |f(re^{i\theta})| &\equiv \frac{1}{2\pi} \frac{R+r}{R-r} \int_0^{2\pi} |n^+ |f(Re^{i\phi})| d\phi \\
 &\leq \frac{1}{2\pi} \frac{R+1}{R-r} \frac{2M}{1-R^2} = \frac{M}{r(R-r)(1-R)} = \frac{4M}{\pi(1-r)^2} \\
 &\leq \frac{2M}{(1-r)^2} .
 \end{aligned}$$

This proves (7.18). ■

Compare Hille[1, page 250(line 22)].

THEOREM 7.11. Given $t_0 \in \mathbb{R}$ and $0 < \delta < 1$. The rectangle $[\frac{1}{2}, \frac{1}{2} + \delta] \times [t_0 - \delta, t_0 + \delta]$ must contain ZEROS of $\varphi_N(s)$ whenever N is sufficiently large. It is understood here that $N < \infty$.

Proof. Take $0 < \delta < \frac{1}{5}t_0$ WLOG and suppose that the theorem is false. We can THEN select a subsequence $\mathcal{J}$ such that:

$$\varphi_N(s) \neq 0 \quad \text{on} \quad \left[\frac{1}{2}, \frac{1}{2} + \delta\right] \times [t_0 - \delta, t_0 + \delta] \quad \text{whenever} \quad N \in \mathcal{J} .$$

Recall that $|\varphi_N(\frac{1}{2} + it)| \equiv 1$ for $t \in \mathbb{R}$. By the Schwarz reflection principle, $\varphi_N(s)$ extends holomorphically to $[\frac{1}{2} - \delta, \frac{1}{2} + \delta] \times [t_0 - \delta, t_0 + \delta]$. In fact:

$$(7.22) \quad \varphi_N(\tfrac{1}{2} - h + it) \overline{\varphi_N(\tfrac{1}{2} + h + it)} \equiv 1 \quad \text{for} \quad 0 \leq h \leq \delta \quad \text{and} \quad |t - t_0| \leq \delta .$$

In this equation $N \in \mathcal{J}$.

By applying Hurwitz's theorem to proposition 7.8, we quickly discover that:

$$\varphi_{\infty}(s) \neq 0 \quad \text{for} \quad s \in (\tfrac{1}{2}, \tfrac{1}{2} + \delta) \times (t_0 - \delta, t_0 + \delta) .$$

Cf. 527(3.1) and Ahlfors[1, p.176]. We also note that

$$\inf_{N \in \mathcal{N}} |\varphi_N(\frac{1}{2} + \frac{\delta}{2} + it_0)| > 0.$$

Take $\varepsilon = \frac{t_0}{2}$ and apply lemma 7.9 to $\varphi_N(s)/\mathcal{C}_6(\varepsilon)$. Cf. 574(7.15). It follows that:

$$\sup_{0 \leq h \leq \frac{\delta}{10}} \int_{t_0 - \frac{\delta}{2}}^{t_0 + \frac{\delta}{2}} \ln \left| \frac{\mathcal{C}_6(\varepsilon)}{\varphi_N(\frac{1}{2} + h + it)} \right| dt = O(1) \quad \text{for } N \in \mathcal{N}.$$

From this estimate we deduce that

$$(7.23) \quad \iint_B |\ln |\varphi_N(s)|| d\sigma dt = O(1)$$

for $N \in \mathcal{N}$ and $B = [\frac{1}{2} - \frac{\delta}{10}, \frac{1}{2} + \frac{\delta}{10}] \times [t_0 - \frac{\delta}{2}, t_0 + \frac{\delta}{2}]$. To check this: we NOTE that

$$\begin{aligned} \iint_B |\ln |\varphi_N(s)|| d\sigma dt &= 2 \iint_{B \cap \{\sigma > \frac{1}{2}\}} |\ln |\varphi_N(s)|| d\sigma dt \\ &= 2 \iint_{B \cap \{\sigma > \frac{1}{2}\}} \left| \ln \frac{|\varphi_N|}{\mathcal{C}_6} + \ln \mathcal{C}_6 \right| d\sigma dt \\ &= O(1) + 2 \int_{\frac{1}{2}}^{\frac{1}{2} + \frac{\delta}{10}} \int_{t_0 - \frac{\delta}{2}}^{t_0 + \frac{\delta}{2}} \ln \frac{\mathcal{C}_6}{|\varphi_N|} dt d\sigma = O(1). \end{aligned}$$

In view of lemma 7.10: $\varphi_N(s) = O(1)$ for $N \in \mathcal{N}$ and $|s - \frac{1}{2} - it_0| \leq \frac{\delta}{15}$.

Proposition 7.8 plus Vitali's theorem will NOW yield:

$$(7.24) \quad \varphi_N(s) \rightrightarrows \varphi_\infty(s) \quad \text{on } |s - \frac{1}{2} - it_0| \leq \frac{\delta}{20} \quad \text{for } N \in \mathcal{N}.$$

Cf. Golusin[1, pp.13-14] and Hille[1, pp.251-255]. By taking absolute values we see that

$$|\varphi_\infty(\frac{1}{2} + it)| \equiv 1 \quad \text{on } |t - t_0| \leq \frac{\delta}{20}.$$

But this contradicts equation (3.1). ■

COROLLARY 7.12. The set $(\frac{1}{2}-\delta, \frac{1}{2}) \times [t_0 - \delta, t_0 + \delta]$ contains poles of $\varphi_N(s)$ whenever N is sufficiently large.

Proof. Trivial. ■

This result corresponds to Selberg[2,page 91 line 5].

We'll return to this line of thought in volume 3. ■

NOTES FOR CHAPTER ELEVEN

1.(page 507 line 19) Cf. Magnus-Karrass-Solitar[1,pp.71-72].

2.(page 507 line 21) Note that this region does not satisfy 211(ℓm). A trivial modification quickly eliminates this artificial difficulty. Cf. 143 line 19 and figure 4.

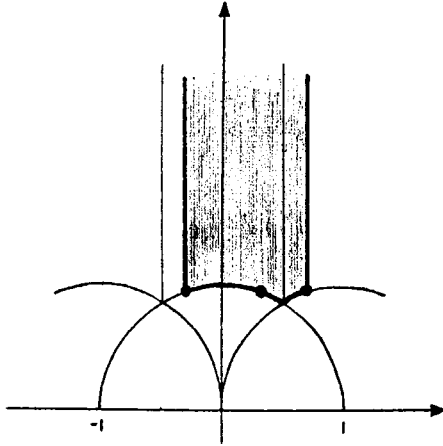


Fig. 4

3.(page 508 line 3) Compare 212 lines 6-8.

4.(page 510 equation 2.7) We stress that the φ_n are CUSP forms when $n > 0$.

4A.(In the interest of clarity) The following *general* remark should also be noted. Consider the situation on page 267. Suppose that $\mathcal{X} = \overline{\mathcal{X}}$. One can *then* assume WLOG that the eigenfunctions φ_n are real λ -vectors.

To check this: note that $\overline{\mathcal{L}_2(\lambda)} = \mathcal{L}_2(\lambda)$ for every $0 \leq \lambda < \infty$. Cf. page 282.

Let $\mathcal{R}(\lambda) = \{f \in \mathcal{L}_2(\lambda) : f \text{ is real}\}$. Use Gram-Schmidt to construct a (real) orthonormal basis for $\mathcal{R}(\lambda)$. Observe that $\mathcal{L}_2(\lambda) = \mathcal{R}(\lambda) + i \mathcal{R}(\lambda)$ as a *direct* sum.

A similar argument shows that the eigenvectors for any unitary symmetric matrix can be chosen to be real. Cf. page 270 top.

5.(page 511 line 3) Recall 458(2.33).

6.(page 511 line 7) $S(T)$ shouldn't be confused with $\pi^{-1}\text{Arg } \int(\frac{1}{2} + iT)$.

7.(page 512 bottom) In recent years there has been much discussion of small eigenvalues. There now exist several (other) ways of showing that $\lambda_1 > \frac{1}{4}$ for $\text{PSL}(2, \mathbb{Z})$. The *underlying* techniques can be divided into three main categories:

- (a) the L_2 method ;
- (b) the triangle-minimax method ;
- (c) the Maass method .

Method (a) has been amply illustrated on pages 512, 528, 529.

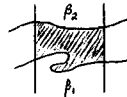
Method (b) is *essentially* the Weyl minimax procedure (*) applied to a NE triangulation. This approach was first used by Atle Selberg around 1951. It was later *rediscovered* by P.Buser[1]. The basic ideas can be gleaned from pages 30-33 of Buser's paper .

The following items should also be mentioned.

I Selberg used the following fact in place of Buser's lemma 1.

Δ LEMMA 1A. Let $\mathcal{R}$ be a compact subset of H bounded by $x=a$ & $x=b$ on the sides AND by Jordan arcs β_j on the top & bottom. Assume that:

- (i) β_j is piecewise smooth ;
- (ii) β_2 is on top ;
- (iii) Ψ is real-valued and C^1 on $\mathcal{R}$;
- (iv) β_1 is decomposable into FINITELY many single-valued functions of x ;
- (v) $\Psi = 0$ along β_1 ;
- (vi) β_2 is expressible as *one* single-valued function of x .



Then:

$$\iint_{\mathcal{R}} |\text{grad } \Psi|^2 dx dy \geq \left(\frac{1}{4} + \frac{1}{16 \sinh^2 \frac{\delta}{2}} \right) \int_{\mathcal{R}} \Psi^2 d\mu(z)$$

where δ is the NE diameter of $\mathcal{R}$.

(*) Cf. Weyl[3] pp.125-127 and Garabedian[1] pp.387,395.

Proof. Let $y = y_j(x)$ represent β_j . The function y_1 may be multi-valued.

In an obvious notation:

$$\begin{aligned}
 \int_{y_1}^{y_2} \psi^2 \frac{dy}{y^2} &= \int_{y_1}^{y_2} \psi^2 d\left(\frac{1}{y_2} - \frac{1}{y}\right) \\
 &= \int_{y_1}^{y_2} \left(\frac{1}{y} - \frac{1}{y_2}\right) 2\psi \frac{\partial \psi}{\partial y} dy \quad \{\text{using integration by parts}\} \\
 &= 2 \int_{y_1}^{y_2} \left(1 - \frac{y}{y_2}\right) \frac{\psi}{y} \frac{\partial \psi}{\partial y} dy \\
 &\leq 2 \left(1 - \frac{\inf y_1}{\sup y_2}\right) \int_{y_1}^{y_2} \frac{|\psi|}{y} \left|\frac{\partial \psi}{\partial y}\right| dy \\
 &\leq 2(1 - e^{-\delta}) \int_{y_1}^{y_2} \frac{|\psi|}{y} \left|\frac{\partial \psi}{\partial y}\right| dy \\
 &\leq 2(1 - e^{-\delta}) \sqrt{\int_{y_1}^{y_2} \psi^2 \frac{dy}{y^2}} \sqrt{\int_{y_1}^{y_2} \left(\frac{\partial \psi}{\partial y}\right)^2 dy}
 \end{aligned}$$

$\Downarrow$

$$\int_{y_1}^{y_2} \psi^2 \frac{dy}{y^2} \leq 4(1 - e^{-\delta})^2 \int_{y_1}^{y_2} \left(\frac{\partial \psi}{\partial y}\right)^2 dy$$

$\Downarrow$

$$\int_{y_1}^{y_2} \left(\frac{\partial \psi}{\partial y}\right)^2 dy \geq \left(\frac{1}{4} + \frac{1}{16 \sinh^2 \frac{\delta}{2}}\right) \int_{y_1}^{y_2} \psi^2 \frac{dy}{y^2}$$

$$\iint_{\mathcal{D}} \left(\frac{\partial \psi}{\partial y}\right)^2 dk dy \geq \left(\frac{1}{4} + \frac{1}{16 \sinh^2 \frac{\delta}{2}}\right) \int_{\mathcal{D}} \psi^2 d\mu.$$

The desired estimate is now manifest. ■

In connection with this result, see page 234 line 10.

[II] Level sets for D -eigenfunctions can be discussed by combining 570(7.6) with an obvious *extension* of Osgood[1] pages 695-696. Compare: Courant-Hilbert[1] page 395(note 7).

[III] Lemma 2 in Buser[1] holds *equally well* when one or more of the vertices lie along ∂H . This fact is easily demonstrated by an approximation argument.

[IV] Let Γ be *any* Fuchsian group as on pages 267 & 338. Take $m=0$, $\% = 1$. Page 33 of Buser[1] is easily extended to this more general setting. Let $\{u_0, u_1, \dots, u_N\}$ be the L_2 eigenfunctions which correspond to $0 \leq \lambda \leq \frac{1}{4}$. Cf. 413 line 6. As usual: $Du_n + \lambda_n u_n = 0$, $0 = \lambda_0 < \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_N \leq \frac{1}{4}$.

Assume that $N \geq 4g+2s+2K-2$. Inequality (12) should now refer to an appropriate linear combination of u_i as in Courant-Hilbert[1,p.406(bottom)]. The triangulation of $\Gamma \setminus H$ is introduced by means of Fricke-Klein[1,pp.318-319] and Keen[1]. We can also take φ_i to be the characteristic function of the i^{th} triangle when $1 \leq i \leq 4g+2s+2K-2$. Note that at least one of these triangles can always be *chosen* to be compact. Cf. Fricke-Klein[1] page 318(lines 17-19) and Keen[1] page 4(line 9), 5(remark), and 14 (re $P(p_0; \mathcal{S})$). The case of a Schwarz Δ group is slightly exceptional and is handled separately.

In any event: we now reach a contradiction. It follows that

$$\lambda_{4g+2s+2K} > \frac{1}{4} \quad .$$

This result corresponds to Selberg[1,p.74 footnote]. Compare Buser[2,pp.44-49] and Zograf[1]. See also page 2k. (★)

▲ 8.(continuation) Similar techniques allow us to prove that $\lambda_1 > \frac{1}{4}$ for ANY triangle group $\Gamma(\nu_1, \nu_2, \nu_3)$. As usual $\frac{1}{\nu_1} + \frac{1}{\nu_2} + \frac{1}{\nu_3} < 1$.

The *idea* of the proof is as follows. Let $\mathcal{F}$ correspond to $T(\frac{\pi}{\nu_1}, \frac{\pi}{\nu_2}, \frac{\pi}{\nu_3})$ followed by *one* reflection. Cf. Magnus[1] pages 65, 81, 86. There is no loss of generality if

(★) The case of arbitrary $\%$ is left to the reader.

we assume that the common side is located along the imaginary axis. We can now extend
 590 note 13. Write $\mathcal{F} = T \cup \tilde{T}$ in an obvious notation. Odd eigenfunctions vanish
 along $\partial T \cup \partial \tilde{T}$. Even eigenfunctions have zero normal derivative instead.

Let f correspond to λ_1 . WLOG f is real-valued.

Review the following facts:

- (i) $\iint_E [|\text{grad } f|^2 + f \Delta f] dx dy = \int_{\partial E} f \frac{\partial f}{\partial n} |dx|$;
- (ii) $\iint_E f^2 \frac{dx dy}{y^2} = 2 \iint_E \frac{f}{y} \frac{\partial f}{\partial y} dx dy + \int_{\partial E} f^2 \frac{dx}{y}$;
- (iii) $\int_E f^2 d\mu \leq 2 \sqrt{\int_E f^2 d\mu} \sqrt{\iint_E \left(\frac{\partial f}{\partial y}\right)^2 dx dy} + \int_{\partial E} f^2 \frac{|dx|}{y}$;
- (iv) $\int_{\mathcal{L}} f^2 \frac{|dx|}{y} \leq \int_{\mathcal{L}} f^2 \frac{|dx|}{y}$;
- (v) $|\text{grad } f|^2 \equiv f_x^2 + f_y^2$;

(vi) one knows what equality means in the Cauchy-Schwarz inequality ;

(vii) page 70 (lines 8-15) and 234 (line 10) ;

(viii) lemma 1A plus Buser[1] lemmas 1 & 2 .

We can now develop some useful estimates for λ_1 by applying (i) - (viii) to
 appropriate subregions of $\mathcal{F}$. These subregions have boundaries which are contained
 in $\partial T \cup \partial \tilde{T} \cup \{f=0\}$. In each case: the *cuspidal* contribution will drop out.
 The exact procedure depends on whether f is odd or even. The case of compact $\mathcal{F}$
 should be used as a guide. Compare Sarnak[1] pages 61-64.

In this way: we conclude that $\lambda_1 > \frac{1}{4}$.

This result was known to Selberg (in the early fifties).

9.(continuation) It remains to consider method (c). To explain the *technique*
 in its simplest form we take $\Gamma = \text{PSL}(2, \mathbb{Z})$. Let $\mathcal{F}$ be the usual fundamental polygon.
 Take $m=0$ & $\% = 1$. Let φ correspond to λ_1 . In view of 580 note 4 :

$$\varphi(z) = \sum_{n \neq 0} c_n y^{\frac{1}{2}} K_{s-\frac{1}{2}}(2\pi n/y) e^{2\pi i n x}, \quad \text{where } \lambda_1 = s(1-s).$$

Note that $\varphi = O(1)$. Let $M = \sup_{z \in H} |\varphi(z)|$. Choose $z_0 \in \mathcal{F}$ so that $|\varphi(z_0)| = M$. Set $z_0 = \xi + iv$ and observe that $v \geq \frac{1}{2}\sqrt{3}$.

For nonzero N we immediately obtain:

$$c_N = \frac{\int_{-\frac{1}{2}}^{\frac{1}{2}} \varphi(x+iy) e^{-2\pi i N x} dx}{y^{\frac{1}{2}} K_{s-\frac{1}{2}}(2\pi N/y)} \quad \{0 < y < \infty\}$$

$$|c_N| \leq \frac{M}{\sup_{0 < y < \infty} y^{\frac{1}{2}} |K_{s-\frac{1}{2}}(2\pi N/y)|}$$

$$|c_N| \leq \frac{M}{\sup_{0 < y < \infty} y^{\frac{1}{2}} |K_{s-\frac{1}{2}}(y)|} \sqrt{2\pi/N}.$$

Therefore:

$$M \leq M \sum_{n \neq 0} \frac{\sqrt{2\pi|n|}}{\sup y^{\frac{1}{2}} |K_{s-\frac{1}{2}}(y)|} |v^{\frac{1}{2}} K_{s-\frac{1}{2}}(2\pi n/v) e^{2\pi i n \xi}|$$

$$M \leq 2M \sum_{n=1}^{\infty} \frac{\sqrt{2\pi n}}{\sup y^{\frac{1}{2}} |K_{s-\frac{1}{2}}(y)|} v^{\frac{1}{2}} |K_{s-\frac{1}{2}}(2\pi n v)|$$

$$\sup_{0 < y < \infty} y^{\frac{1}{2}} |K_{s-\frac{1}{2}}(y)| \leq 2 \sum_{n=1}^{\infty} \sqrt{2\pi n v} |K_{s-\frac{1}{2}}(2\pi n v)|.$$

Suppose (for one moment) that $\frac{1}{2} \leq s \leq 1$. By applying page 22(iii), we quickly establish that

$$K_0(x) \leq K_{s-\frac{1}{2}}(x) \leq K_{\frac{1}{2}}(x)$$

for $x > 0$. In addition:

$$K_{\frac{1}{2}}(x) = \sqrt{\frac{\pi}{2x}} e^{-x}.$$

Cf. Watson[1] page 80. It follows that:

$$y^{\frac{1}{2}} K_0(y) \leq 2 \sum_{n=1}^{\infty} \sqrt{2\pi n y} \sqrt{\frac{\pi}{2 \cdot 2\pi n y}} e^{-2\pi n y}$$

$$y^{\frac{1}{2}} K_0(y) \leq \sqrt{2\pi} \sum_{n=1}^{\infty} e^{-2\pi n y}$$

$$y^{\frac{1}{2}} K_0(y) \leq \sqrt{2\pi} \frac{e^{-2\pi y}}{1 - e^{-2\pi y}}$$

$$y^{\frac{1}{2}} K_0(y) \leq \sqrt{2\pi} \frac{1}{e^{2\pi y} - 1}$$

$$\frac{y^{\frac{1}{2}} K_0(y)}{\sqrt{2\pi}} \leq \frac{1}{e^{2\pi y} - 1}$$

for ANY $y > 0$. Take $y = \frac{1}{5}$ (say). This yields an *immediate* contradiction since

$$\frac{y^{\frac{1}{2}} K_0(y)}{\sqrt{2\pi}} = .31270412 \quad \text{and} \quad \frac{1}{e^{2\pi y} - 1} = .00435228.$$

Cf. Watson[1] page 698 or Abramowitz-Stegun[1] page 417. We conclude that $\lambda_1 > \frac{1}{4}$.

To improve this result: take $s = \frac{1}{2} + iR$ where $R > 0$. By 590 note 13 there is no loss of generality if we assume that φ is either even or odd. Parseval's equation will *now* yield:

$$\int_{-\frac{1}{2}}^{\frac{1}{2}} |\varphi(x+iy)|^2 dx = \sum_{n \neq 0} |c_n|^2 |y^{\frac{1}{2}} K_{s-\frac{1}{2}}(2\pi n|y)|^2 = 2 \sum_{n=1}^{\infty} |c_n|^2 |y^{\frac{1}{2}} K_{s-\frac{1}{2}}(2\pi n y)|^2$$

$$\left\{ \text{since } |c_{-n}| = |c_n| \right\}$$

$$|c_N| \leq \frac{1}{\sqrt{2}} \frac{M}{\sup_{0 < y < \infty} y^{\frac{1}{2}} |K_{s-\frac{1}{2}}(2\pi n|y)|}$$

$$|c_N| \leq \frac{1}{\sqrt{2}} \frac{M}{\sup_{0 < y < \infty} y^{\frac{1}{2}} |K_{s-\frac{1}{2}}(y)|} \sqrt{2\pi n|y|} .$$

Therefore:

$$\sup_{0 < y < \infty} y^{\frac{1}{2}} |K_{s-\frac{1}{2}}(y)| \leq \sqrt{2} \sum_{n=1}^{\infty} \sqrt{2\pi n y} |K_{s-\frac{1}{2}}(2\pi n y)| .$$

By applying 22(iii) once more, we see that

$$|K_{iR}(x)| \leq \int_0^{\infty} e^{-x \cosh t} dt = K_0(x) \leq K_{\frac{1}{2}}(x) \quad \text{for } 0 < x < \infty .$$

It follows that

$$y^{\frac{1}{2}} |K_{iR}(y)| \leq \sqrt{\pi} \frac{e^{-2\pi y}}{1 - e^{-2\pi y}}$$

$$\frac{y^{\frac{1}{2}} |K_{iR}(y)|}{\sqrt{\pi}} \leq \frac{1}{e^{2\pi y} - 1} \leq \frac{1}{e^{\pi\sqrt{3}} - 1}$$

for ANY $y > 0$. Take $y = \frac{1}{5}$ again and use a few seconds of computer time to graph $K_{iR}(\frac{1}{5})$. Cf. Hejhal[4] page 102(bottom). This procedure yields $R > 1.4$.

In other words: $\lambda_1 > 2.21$.

These estimates can *obviously* be improved. They do however serve to illustrate Maass' method.

Our discussion should be compared to Maass[1] pages 155-156. Note that Maass attributes the *idea* of this method to C.L.Siegel.

10.(continuation) For further examples: see note 77 .

11.(continuation) Take $\Gamma = \text{PSL}(2, \mathbb{Z})$ as in note 9. The Weyl minimax procedure allows us to determine an *upper* bound for λ_1 .

To accomplish this let $\{\mu_n\}_{n=1}^{\infty}$ denote the *odd* eigenvalues for $\Gamma \backslash H$. Cf. page 513 and 590 note 13. The minimax procedure is applied to $\mathcal{F}$ in the following form. Let $v_1, \dots, v_{n-1}$ be any L_2 functions on $\mathcal{F}$. Put

$$d[v_1, \dots, v_{n-1}] = \inf \frac{\iint_{\mathcal{F}} |\text{grad } \psi|^2 dx dy}{\int_{\mathcal{F}} \psi^2 d\mu}$$

where ψ ranges over *all* piecewise smooth CONTINUOUS functions on $\mathcal{F}$ which

- (a) are *odd* with respect to x ;
- (b) vanish near $\partial\mathcal{F}$;
- (c) have compact support ;
- (d) are orthogonal to $v_1, \dots, v_{n-1}$. (*)

Then:

$$\mu_n = \max_{\mathcal{F}} d[v_1, \dots, v_{n-1}] .$$

The proof is similar to Courant-Hilbert[1] pages 406-407. Page 244(2.5) is used to

compute $\iint_{\mathcal{F}} |\text{grad } \psi|^2 dx dy$ when ψ is C^2 . One finds that

(*) v & ψ real

$$h = \sum_{\sigma} c_m \varphi_m \quad \Psi = \sum_{\sigma} \frac{c_m}{a(1-a) - \lambda_m} \varphi_m \quad (\star)$$

$$\iint_{\mathcal{F}} |\text{grad } \Psi|^2 dx dy = \iint_{\mathcal{F}} \Psi (-\Delta \Psi) dx dy = \sum_{\sigma} \frac{\lambda_m c_m^2}{[a(1-a) - \lambda_m]^2}$$

since $E(z; \frac{1}{2} + it)$ is even. See also the 2-dimensional analog of 220 top.

A similar minimax problem holds for $\mathcal{R} = [-\frac{1}{2}, \frac{1}{2}] \times [1, \infty)$. Cf. Titchmarsh[1] pages 22, 49, 110. By extending Ψ to be zero on $\mathcal{F} - \mathcal{R}$ we discover that

$$\mu_n(\mathcal{F}) \leq \mu_n(\mathcal{R}) .$$

Cf. Courant-Hilbert[1] pages 408-409.

Let $\mu = \frac{1}{4} + \rho^2$. The computation of $\mu_n(\mathcal{R})$ leads to $K_{i\rho}(2\pi N) = 0$ as in Hejhal[4] page 97. We conclude that

$$\begin{aligned} \rho_1(\mathcal{F}) &\leq \rho_1(\mathcal{R}) = 9.76877 \\ \rho_2(\mathcal{F}) &\leq \rho_2(\mathcal{R}) = 12.4485 \\ \rho_3(\mathcal{F}) &\leq \rho_3(\mathcal{R}) = 14.6850 \\ \rho_4(\mathcal{F}) &\leq \rho_4(\mathcal{R}) = 16.6916 \\ \rho_5(\mathcal{F}) &\leq \rho_5(\mathcal{R}) = 16.9297 \\ \rho_6(\mathcal{F}) &\leq \rho_6(\mathcal{R}) = 18.5494 \\ \rho_7(\mathcal{F}) &\leq \rho_7(\mathcal{R}) = 20.2544 \\ \rho_8(\mathcal{F}) &\leq \rho_8(\mathcal{R}) = 20.2999 . \end{aligned}$$

These estimates should be compared with 513(top) and Roelcke[3] page 52. At the same time:

note that $\rho_n(\mathcal{F}) \geq \rho_n(\mathcal{R}_0)$ where $\mathcal{R}_0 = [-\frac{1}{2}, \frac{1}{2}] \times [\frac{1}{2}\sqrt{3}, \infty)$. Cf.

Hejhal[4] pages 106-107.

The even analog is irrelevant because (there) the continuous spectrum appears.

($\star$) σ means restrict to odd eigenfunctions

12.(continuation) The emphasis in chapter 11 is on classical groups. Cf. page 501 IV and 507 top. Readers interested in *small* eigenvalues for more general groups are referred to Buser[1-5], Huber[1-4], Randol[4], Schoen[1], and Schoen-Wolpert-Yau[1]. Compare: note 67 and Hejhal[6].

13.(page 513 line 9) To justify this dichotomy we remark that $R[f] = f(-x+iy)$ is a self-adjoint operator on $L_2(\Gamma \backslash H)$ which commutes with D . We can now construct a simultaneous diagonalization. Cf. 228 note 142 and 262 line 8.

Compare Maass[2,p.260] and Petersson[11,pp.410-411].

14.(page 513 line 10) It is commonly conjectured that the eigenvalues of $PSL(2, \mathbb{Z})$ are all simple. This conjecture has never been proved. The corresponding assertion for congruence subgroups is definitely false; cf. Randol[3].

There are similar questions for $\mathcal{J}(s)$. Cf. Brent[1], Edwards[1,pp.171-181], Heath-Brown[1], Lehmer[1], Montgomery[2], and Titchmarsh[4,pp.322,329].

15.(page 514 line 13) Compare Venkov[6].

16.(page 514 line 13) One should also consider Colin de Verdiere[2,3].

17.(page 514 line 16) Selberg developed an *equivalent* approach around 1953. In this connection [▲]the reader may wish to consult:

- | | |
|--|---------------------------------|
| (a) Selberg[1,p.85(line 4)] | (e) Vignéras[1,pp.95,96,98,129] |
| (b) Eichler[3] chapter IV | (f) Andrianov-Fomenko[1] |
| (c) Eichler[4] pages 100-103 and 130-133 | (g) Kuznecov[3] (★) |
| (d) Landau[1] Sätze 209, 214, 897 | (h) Golovčanskiĭ-Smotrov[1] . |

18.(page 514 equation 2.16) This S shouldn't be confused $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$.

19.(page 514 equation 2.17) Compare Fricke-Klein[1,p.14] and Hejhal[11,12] .

20.(page 515 lemma 2.2) It is best to establish lemma 2.3 before proving assertion (c).

21.(page 516 bottom) Compare Chowla-Cowles-Cowles[1,p.374(line 21)] .

22.(page 517 equation 2.20) Note that $n(-T) = n(T)$.

23.(page 518 lemma 2.7) Compare Chowla-Cowles-Cowles[1].

24.(page 519 line 9) Among other things: Sarnak shows that

(★) See also Zagier[1] page 304 line 20.

$$\frac{1}{N[d: \varepsilon_d \leq x]} \sum_{\varepsilon_d \leq x} h(d) = \frac{16}{35} \frac{h(x^2)}{x} + O(x^{\frac{2}{5} + \varepsilon}) \quad \forall \varepsilon > 0.$$

25.(page 519 equation 2.22) x^α can *presumably* be replaced by $x^{3/4(\ln x)^A}$.

Cf. Hejhal[13] and Vinogradov-Takhtajan[1].

26.(continuation) By applying equations (2.23) and (2.24), we see that:

$$\sum_{d \leq x} h(d) \ln \varepsilon_d = \frac{\pi^2}{185(3)} x^{3/2} - \frac{3}{\pi^2} x \ln x + (\text{constant})x + O(x^{\frac{1}{2} - \varepsilon}).$$

If we assume the GRH, $x^{\frac{1}{2} - \varepsilon}$ can be replaced by $x^{\frac{1}{2}} \ln \ln x$. Cf. Littlewood[1] and Shanks[1].

27.(page 520 line 6) Cf. Davenport[1,p.40] and Hua[2,p.159] regarding real primitive characters.

28.(page 520 line 17) In chapter 10.

29.(page 520 line 19) Note that $R = Ak(\frac{\pi}{3})A^{-1}$ where $A = \begin{pmatrix} \sqrt{3} & -1 \\ 0 & 2 \end{pmatrix}$.

30.(page 521 line 12) Cf. Siegel[6].

31.(page 522 top) To achieve the proper perspective with regard to multiplier systems on $SL(2, \mathbb{Z})$, we refer to:

- | | |
|-------------------------------------|--|
| (a) Knopp[1] chapter 4 | (f) Rademacher[2] chapter 9 |
| (b) Lehner[1,pp.343,346-350] | (g) Rademacher-Zuckerman[1] |
| (c) Maass[2,pp.132-135,144-146] (*) | (h) Riemann[1]pp.466-478(appendix by Dedekind) |
| (d) Petersson[12] section 1 | (i) Van Lint[1] |
| (e) Rademacher[1] | (j) Weber[1,pp.124-130] . |

32.(page 522 line 18) Use 506(39) and 580(note 2).

33.(page 523 equation 2.26) This agrees with Maass[2,p.144] and Rankin[1,p.208].

(*) after correcting a serious error on page 133 bottom

34.(page 524 line 12) There is an obvious analog of note 2. Cf. figure 5. The reader should also consult 211(8), 421(28), and 423(30).

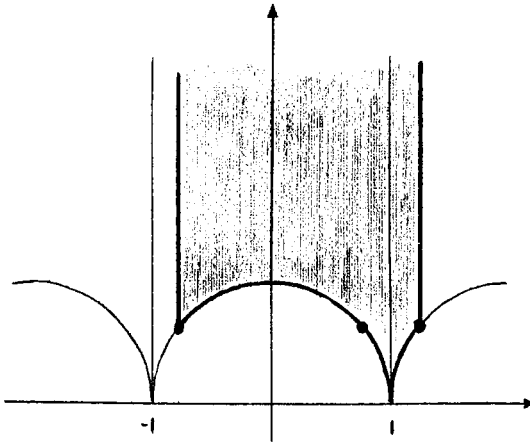


Fig. 5

35.(page 525 line 4) Cf. 212 lines 6-8.

36.(page 526 line 7) Cf. note 34.

37.(page 528 line 6) Cf. note 4.

38.(page 529 equation 3.8) This equation agrees with Atkin-Lehner[1,lemma 9]. Cf. 525 line 1 and observe that $RAR^{-1} = \frac{1}{\sqrt{2}} \begin{pmatrix} -2 & -1 \\ 6 & 2 \end{pmatrix}$.

39.(page 530 line 15) Cf. note 34 in connection with 426(64).

40.(page 530 bottom) To achieve the proper perspective concerning multiplier systems on $\overline{\Gamma}$, we refer to:

(a) 333 line 3

(b) 525 line 4

(c) Berndt[1] theorem 4.1

(d) Fricke[1] pp.491-494

(e) Knopp[1] chapter 4

(f) Rademacher[4]

(g) Shimura[3] page 447(1.10)

(h) Weber[1] sections 21,24,25,34,39,40 .

- 41.(page 532 line 7) Cf. also 505(33).
- 42.(page 532 line 9) In chapter 9.
- 43.(page 532 line 11) The *cuspidal* computations in sections 4,5,6 date back to 1974.
- 44.(page 532 line 16) The extension of our results to the case of arbitrary N seems like a worthwhile project.
- 45.(page 533 line 3) Cf. Chuman[1] and Rademacher[3] regarding the generators of $\hat{\Gamma}_0(N)$.
- 46.(continuation) For some pictures of fundamental regions, we refer to Birch[1] and Fricke[2,3].
- 47.(page 533 line 13) The reader is urged to consult 211(8), 327(11), 421(28), 423(30), 426(64), and 506(39).
- 48.(page 534 lemma 4.4) As a DISJOINT union of double cosets.
- 49.(page 535 lemma 4.5) Cf. note 47.
- 50.(page 538 line 15) And 505(33).
- 51.(page 540 top) We emphasize that $m \equiv 0 \pmod{2}$.
- 52.(page 540 line 7) See also 505(note 33).
- 53.(page 540 line 13) Cf. note 44.
- 54.(page 540 line 16) Note that $\hat{\Gamma}(N) \cong \Gamma(N)$ for $N \geq 3$.
- 55.(page 541 line 5) Cf. Rankin[1,p.46(2.2.24)] regarding $n(z, \Gamma)$.
- 56.(page 541 line 7) Compare Frasch[1]. For pictures of fundamental regions, consult Klein-Fricke[1] pages 222,279,354,355,367,373,398.
- 57.(page 541 line 8) Cf. note 47.
- 58.(page 542 lemma 5.1) Cf. note 47.
- 59.(page 544 line 8) Cf. note 54.
- 60.(page 545 line 5) These p_j have nothing to do with 533 line 2.
- 61.(page 547 line 6) There should be no confusion here with 338(iii).
- 62.(page 548 line 3) Note that $M \neq 1$.
- 63.(page 548 line 4) Remember that $N = KM$.
- 64.(page 549 line 2) Cf. Orihara[1] for arbitrary N .

65.(page 549 line 3) We stress that $N \neq 1$.

66.(page 549 line 18) Take $m = 0$ and $\mathcal{W} = 1$. By combining Orihara[1], theorem 5.12, 327(15), and 328(21), we quickly establish that

$$M_e = 0 \quad \text{whenever} \quad \hat{\Gamma}(N) \subseteq \Gamma \subseteq \text{PSL}(2, \mathbb{Z}) \quad .$$

NOTE that there is no restriction on N .

67.(continuation) For appropriate *noncongruence* subgroups of $\text{PSL}(2, \mathbb{Z})$ the number M_e can be made arbitrarily large. $\blacktriangle$ Cf. Selberg[5] and 735(note 16A).

68.(page 550 line 4) Don't confuse $\mathcal{K}$ and K . This equation has $\mathcal{K}$.

69.(page 552 line 2) These manipulations *do* require some thought. Cf. lemma 5.14.

70.(page 552 equation 5.16) As usual $K = NM^{-1}$.

71.(page 553 proposition 5.15) Cf. note 65.

72.(page 554 line 3) Some readers may want to go further and actually verify that

$$\Phi(s)\Phi(1-s) = I \quad .$$

73.(page 558 line 10) Cf. note 68.

74.(page 563 line 3) This equation requires a little thought.

75.(page 565 line 8) Cf. note 65.

76.(page 566 line 3) Fortunately we were able to *simplify* our original proof of theorem 5.21. It no longer requires 250 pages. On the other hand: note that section 5 is rather condensed.

77.(page 566 line 4) Take $m=0$ and $\mathcal{W}=1$. It is commonly conjectured that $\lambda_1 \geq \frac{1}{4}$ for *all* N . This conjecture has never been proved. $\blacktriangle$ Cf. Selberg[5]pages 13-14.

In this note we *show* that $\lambda_1 > \frac{1}{4}$ for $2 \leq N \leq 5$.

To begin with: review notes 4A, 13, 66. Let φ correspond to $\lambda_1 = s(1-s)$.

WLOG φ is real-valued. For purposes of contradiction: *assume* that $\frac{1}{2} \leq s < 1$.

Let $\mathcal{F}_1$ be the usual fundamental polygon for $\text{PSL}(2, \mathbb{Z})$. Write

$$\text{PSL}(2, \mathbb{Z}) = \sum_{j=1}^{\infty} \hat{\Gamma}(N) L_j$$

with $L_1 = I$. Recall that $\hat{\Gamma}(N)$ is a normal subgroup of $\text{PSL}(2, \mathbb{Z})$.

The following region can (obviously) be used as a fundamental region for $\hat{\Gamma}(N)$:

$$\mathcal{F} = \bigcup_{j=1}^R L_j(\mathcal{F}_1) .$$

Let $M = \sup_{z \in \mathcal{H}} |\varphi(z)|$. Since $\varphi(L_j z)$ also belongs to $\mathcal{L}_2(\lambda_1)$ [cf. page 282] there is no loss of generality if we assume that $|\varphi(z_0)| = M$ for some point $z_0 \in \mathcal{F}_1$.

We can now try to imitate note 9.

In this case:

$$\varphi(z) = \sum_{n \neq 0} c_n Y^{\frac{1}{2}} K_{s-\frac{1}{2}}(2\pi|n|Y) e^{2\pi i n X}$$

where $Z = X + iY = \frac{z}{N}$. Cf. pages 271 and 282(3.9). Since φ is real, $c_{-n} = \overline{c_n}$.

Let $Z_0 = X_0 + iY_0 = \frac{z_0}{N}$. By imitating page 587(top) we see that

$$\sup_{0 < y < \infty} y^{\frac{1}{2}} |K_{s-\frac{1}{2}}(y)| \leq \sqrt{2} \sum_{n=1}^{\infty} \sqrt{2\pi n Y_0} |K_{s-\frac{1}{2}}(2\pi n Y_0)|$$

$$y^{\frac{1}{2}} K_{s-\frac{1}{2}}(y) \leq \frac{\sqrt{\pi}}{e^{2\pi Y_0} - 1}$$

$$y^{\frac{1}{2}} K_0(y) \leq \frac{\sqrt{\pi}}{e^{\frac{\pi}{N} \sqrt{3}} - 1} \quad (y > 0) .$$

The following table takes care of $2 \leq N \leq 4$:

N	y	$y^{\frac{1}{2}} K_0(y)$	$\sqrt{\pi} (e^{\frac{\pi}{N} \sqrt{3}} - 1)^{-1}$
2	0.2	.78383299	.12490040
3	0.2	.78383299	.34525806
4	0.2	.78383299	.61170664
5	0.2	.78383299	.90010584

The case $N = 5$ is considerably more involved. It is necessary to use group representations.

Let $\varphi_1, \dots, \varphi_A$ be an orthonormal basis for $A_0[s(1-s)]$. For each $L \in \text{PSL}(2, \mathbb{Z})$ we have

$$\begin{pmatrix} \varphi_1(Lz) \\ \vdots \\ \varphi_A(Lz) \end{pmatrix} = \mathcal{M}(L) \begin{pmatrix} \varphi_1(z) \\ \vdots \\ \varphi_A(z) \end{pmatrix}$$

where $\mathcal{M}(L)$ is some unitary representation [of degree A] whose kernel includes $\hat{\Gamma}(5)$. The representation $\mathcal{M}$ may be regarded as a unitary representation on $\hat{\Gamma}(1)/\hat{\Gamma}(5)$.

The irreducible representations of $\hat{\Gamma}(1)/\hat{\Gamma}(5)$ are well known. Cf. Hecke[1] page 529 (especially the character table) and Murnaghan[1] page 176. The associated degrees are 1, 3, 3, 4, and 5. Note that:

$$60 = [\hat{\Gamma}(1) : \hat{\Gamma}(5)] = 1^2 + 3^2 + 3^2 + 4^2 + 5^2.$$

Decompose $\mathcal{M}$. There is no loss of generality if we now let $\{\varphi_1, \dots, \varphi_B\}$ correspond to one of the irreducible representations of $\hat{\Gamma}(1)/\hat{\Gamma}(5)$:

$$\begin{pmatrix} \varphi_1(Lz) \\ \vdots \\ \varphi_B(Lz) \end{pmatrix} = \mathcal{M}_0(L) \begin{pmatrix} \varphi_1(z) \\ \vdots \\ \varphi_B(z) \end{pmatrix} \quad (\star)$$

By conjugation we may also assume that $\mathcal{M}_0(S)$ is diagonal.

Let $\rho = e^{2\pi i/5}$. Note that $\rho^4 + \rho^3 + \rho^2 + \rho + 1 = 0$ and that $x^4 + x^3 + x^2 + x + 1$ is irreducible over $\mathbb{Q}$. By referring to the character table in Hecke, we discover that:

($\star$) $B = 1, 3, 4, 5$

$$\mathcal{M}_0(s) = 1 \quad \text{for degree 1}$$

$$\mathcal{M}_0(s) = \begin{pmatrix} \rho & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \rho^4 \end{pmatrix} \quad \text{for degree } 3_1$$

$$\mathcal{M}_0(s) = \begin{pmatrix} \rho^2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \rho^3 \end{pmatrix} \quad \text{for degree } 3_2$$

modulo
permutations

$$\mathcal{M}_0(s) = \begin{pmatrix} \rho & 0 & 0 & 0 \\ 0 & \rho^2 & 0 & 0 \\ 0 & 0 & \rho^3 & 0 \\ 0 & 0 & 0 & \rho^4 \end{pmatrix} \quad \text{for degree 4}$$

$$\mathcal{M}_0(s) = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & \rho & 0 & 0 & 0 \\ 0 & 0 & \rho^2 & 0 & 0 \\ 0 & 0 & 0 & \rho^3 & 0 \\ 0 & 0 & 0 & 0 & \rho^4 \end{pmatrix} \quad \text{for degree 5} \quad .$$

The essential point is that $\mathcal{M}_0(s)^5 = I$.

The function $\sum_{j=1}^B |\varphi_j(z)|^2$ is invariant under $\text{PSL}(2, \mathbb{Z})$. Let its maximum value M^2 be assumed at $z_0 \in \mathcal{F}_1$. Write $|\varphi_j(z_0)| = M L_j$. Then:

$$L_1^2 + L_2^2 + \dots + L_B^2 = 1$$

$$\varphi_j(z) = \sum_{n \neq 0} c_n^{(j)} y^{1/2} K_{s-\frac{1}{2}}(2\pi |n| y) e^{2\pi i n x}$$

$$\sum_{j=1}^B \int_{-\frac{1}{2}}^{\frac{1}{2}} |\varphi_j(z)|^2 dX = \sum_{j=1}^B \sum_{n \neq 0} |c_n^{(j)}|^2 |y^{1/2} K_{s-\frac{1}{2}}(2\pi |n| y)|^2 \leq M^2$$

$$|c_n^{(j)}| \leq \frac{M}{\sup_{0 < y < \infty} y^{\frac{1}{2}} |K_{s-\frac{1}{2}}(y)|} \sqrt{2\pi |n|} \quad .$$

The analog of 585(line 11) is

$$L_j \sup_{0 < y < \infty} y^{\frac{1}{2}} |K_{s-\frac{1}{2}}(y)| \leq \sum_{n \neq 0}^* \sqrt{2\pi \ln Y_0} |K_{s-\frac{1}{2}}(2\pi \ln Y_0)|$$

where * indicates summation over the following residue classes

$$\varphi_j(z+1) \equiv \varphi_j(z) \Rightarrow n \equiv 0 \pmod{5}$$

$$\varphi_j(z+1) \equiv \rho \varphi_j(z) \Rightarrow n \equiv 1 \pmod{5}$$

$$\varphi_j(z+1) \equiv \rho^2 \varphi_j(z) \Rightarrow n \equiv 2 \pmod{5}$$

$$\varphi_j(z+1) \equiv \rho^3 \varphi_j(z) \Rightarrow n \equiv 3 \pmod{5}$$

$$\varphi_j(z+1) \equiv \rho^4 \varphi_j(z) \Rightarrow n \equiv 4 \pmod{5} .$$

It follows that

$$L_j y^{\frac{1}{2}} K_0(y) \leq \sqrt{\frac{\pi}{2}} \sum_{n \neq 0}^* e^{-2\pi \ln Y_0} .$$

By taking $y = 0.2$ we obtain

$$L_j(.78383299) \leq \sqrt{\frac{\pi}{2}} \sum_{n \neq 0}^* e^{-\frac{\pi}{5} \ln \sqrt{3}} .$$

There are now 5 possibilities:

$$\varphi_j(z+1) \equiv \varphi_j(z) \Rightarrow L_j(.78383299) \leq .01090955 \Rightarrow L_j^2 \leq .00019372$$

$$\varphi_j(z+1) \equiv \rho \varphi_j(z) \Rightarrow L_j(.78383299) \leq .44014371 \Rightarrow L_j^2 \leq .31531304$$

$$\varphi_j(z+1) \equiv \rho^2 \varphi_j(z) \Rightarrow L_j(.78383299) \leq .19087247 \Rightarrow L_j^2 \leq .05929793$$

$$\varphi_j(z+1) \equiv \rho^3 \varphi_j(z) \Rightarrow L_j(.78383299) \leq .19087247 \Rightarrow L_j^2 \leq .05929793$$

$$\varphi_j(z+1) \equiv \rho^4 \varphi_j(z) \Rightarrow L_j(.78383299) \leq .44014371 \Rightarrow L_j^2 \leq .31531304 .$$

Review page 597. Since $L_1^2 + \dots + L_B^2$ is less than $\frac{9}{10}$ we have reached the desired contradiction.

This proof should be compared to Maass[1] pages 157-158.

78.(page 566 line 7) Influenced by Shimura[3] page 446 bottom.

79.(page 566 line 11) Note that $\hat{\Gamma}_1(N) \cong \Gamma_1(N)$ for $N \geq 3$.

80.(page 566 equation 6.3) As a DISJOINT union.

81.(page 567 equation 6.7) Note that $\Gamma_{12}(N) = \Gamma_0(N)$ when $N = 3$.

82.(page 567 line 5) We provide a few pointers:

(A) take $T_{xA} = W_{xA} S_{W_{xA}}^{-1}$ in the analog of lemma 5.1;

(B) to state the analogs of lemmas 5.2 & 5.3 simply delete the congruence conditions involving $\tilde{x}_2 - \tilde{x}_1 - (j_2 - j_1)\tilde{x}_1\tilde{x}_2$;

(C) take $0 \leq \delta < A_2|\gamma|$ in the analog of lemma 5.6.

83.(page 568 line 3) It is instructive to check that theorem 6.2 agrees with (4.6) when $N = 3$. Cf. note 81.

84.(page 568 line 12) The notation $G(2\cos\frac{\pi}{N})$ is also used.

85.(page 569 line 2) There is an obvious analog of note 2.

86.(page 569 line 14) Note that $\lambda_1 > \frac{1}{4}$. Cf. page 583 (8).

87.(page 571 line 1) Assertion (b) will now refer to just f_z and $f_{\bar{z}}$.

88.(page 575 equation 7.17) To be more precise: let

$$B_h = \{z \in \mathbb{C} : |x - \frac{1}{2}| \leq \frac{1}{2} - h, |y| \leq 1 - 2h\}.$$

Note that $g(z; \frac{1}{2}; h) \geq g(z; \frac{1}{2}; B_h)$ for $z \in B_h$. It follows that

$$\frac{\partial g}{\partial x}[h + iy; \frac{1}{2}; h] \geq \frac{\partial g}{\partial x}[h + iy; \frac{1}{2}; B_h]$$

for $|y| < 1 - 2h$. But $g(z; \frac{1}{2}; B_h)$ has an OBVIOUS connection with $g(z; \frac{1}{2}; 0)$.

See also Osgood[1] pages 437-440 and 717-722.

89.(page 576 line 7) Compare Tsuji[1] chapters I and

90.(page 579 line 5) See also: Venkov[7,8] .